

CSE / OPEN RESEARCH PROJECT

Knowledge Graph for Indian Scientific Literature

Create a bilingual discovery prototype that links Indian research papers, methods, datasets and institutions while preserving source provenance.

TEAM	DURATION	FORMAT
Up to 4 students	14-week prototype	Mentored research

Problem

Research discovery is hindered by inconsistent terminology, fragmented metadata and limited cross-language search. Generated summaries can also obscure provenance if citations are not explicit.

Method

The team will ingest openly licensed metadata, define a compact ontology, extract entities and relations, evaluate Hindi-English retrieval and expose evidence-backed graph paths. No paywalled full text will be copied.

Research objectives

- Define a provenance-first research ontology
- Build entity resolution and citation-linking baselines
- Evaluate bilingual retrieval with a documented test set
- Expose evidence for every generated relationship

Skills and preparation

- Python and NLP
- Graph databases
- Information retrieval
- Data licensing and evaluation

Expected deliverables

- Ontology and data dictionary
- Curated evaluation dataset
- Search and graph prototype
- Model card, provenance report and demo

Indicative timeline

Weeks 1-3	licensing audit and ontology
Weeks 4-7	ingestion and entity resolution
Weeks 8-11	bilingual retrieval
Weeks 12-14	evaluation and demonstration

Responsible-use note

This brief describes a staging research concept. It does not promise funding, credit, authorship, laboratory access, safety clearance or selection. The faculty lead must approve the final protocol, data access, ethics and safety requirements before work begins.

Apply through the authenticated NSUT Connect project page. Do not send passwords, secret keys, medical information or confidential research data with an expression of interest.